

MATHEMATICS — MID-TERM EXAMINATION 2025

BARCODE
QP-MATH-2025

B.Tech (CSE / ECE) — Semester IV | Time: 3 Hours | Maximum Marks: 100

GENERAL INSTRUCTIONS

- 1. Answer all questions. Each question carries marks as indicated.
- 2. Sub-parts must be answered in order. Partial answers will be awarded partial marks.
- 3. Diagrams must be neat and labelled. Unlabelled diagrams will not receive full marks.
- 4. Use of calculators, mobile phones, or any electronic device is strictly prohibited.
- 5. Write your Roll No. and Barcode Number clearly on the answer sheet.

Q1 [Total: 12 Marks]

Sub-part	Marks	Question
Q1(i)	3	Define a computer system. List its four main functional components with a brief explanation of each.
Q1(ii)	3	Differentiate between primary memory (RAM) and secondary memory (Hard Disk). Give two examples of each.
Q1(iii)	3	What is an operating system? Describe any three functions of an operating system.
Q1(iv)	3	Explain the concept of multiprogramming and multitasking with suitable examples.

Q2 [Total: 12 Marks]

Sub-part	Marks	Question
Q2(i)	4	Draw a neat, labelled diagram of the Von Neumann architecture. Explain the role of each unit.
Q2(ii)	4	Describe the fetch-decode-execute cycle in detail with the help of a flowchart.
Q2(iii)	4	What is cache memory? Explain its working principle and why it improves system performance.

Q3 [Total: 11 Marks]

Sub-part	Marks	Question
Q3(i)	3	Convert the decimal number 255 into binary, octal, and hexadecimal representations.
Q3(ii)	3	Explain the 2's complement representation of negative numbers with an example using 8-bit binary.
Q3(iii)	2	What is the IEEE 754 floating point standard? Represent -6.75 in single precision format.
Q3(iv)	3	Perform binary addition: 01101010 + 00110101. Show all carry steps.

Q4 [Total: 15 Marks]

Sub-part	Marks	Question
Q4(i)	5	Explain the classification of computers based on size and purpose. Give two examples for each category.
Q4(ii)	5	Describe the concept of pipelining in CPU design. What are the stages of a 5-stage pipeline?
Q4(iii)	5	What is virtual memory? Explain demand paging and page replacement algorithms (LRU, FIFO) with examples.

Q5 [Total: 8 Marks]

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Sub-part	Marks	Question
Q5(i)	2	What is ASCII? How is the character 'A' represented in ASCII and Unicode?
Q5(ii)	2	Differentiate between RISC and CISC processor architectures.
Q5(iii)	2	Explain the role of the Arithmetic Logic Unit (ALU) in CPU operations.
Q5(iv)	2	What is bus topology? List its advantages and disadvantages.

Q6 [Total: 12 Marks]

Sub-part	Marks	Question
Q6(i)	4	Write short notes on: (a) CPU registers (b) Instruction Set Architecture
Q6(ii)	4	Explain the working of DMA (Direct Memory Access) with a block diagram.
Q6(iii)	4	Describe different types of ROM (PROM, EPROM, EEPROM) with their characteristics.

Q7 [Total: 10 Marks]

Sub-part	Marks	Question
Q7(i)	5	Differentiate between static RAM (SRAM) and dynamic RAM (DRAM) with respect to speed, cost, and application.
Q7(ii)	5	Explain magnetic disk storage: track, sector, cylinder, and how data is accessed. Include a labelled diagram.